

Waves & Sound - MythBusters Critical-Thinking Worksheets

Season/episode numbering follows Apple TV listings. MythBusters is packaged differently by some broadcasters, so numbering can vary by service. Use only a school/licensed/legal source for playback.

How to use: Do not watch passively. Commit to a prediction first, collect evidence while watching, then explain the physics and evaluate the quality of the test.

1. Breaking Glass

Season 3, Episode 13 (2005) | **Physics:** Resonance, natural frequency, frequency, amplitude and energy transfer by sound

A singer tries to shatter a glass using sound. It gives students a concrete way to connect vibration, frequency and resonance.

BEFORE - predict

1. What must be true about the sound frequency if the glass is to vibrate strongly?

2. Is singing louder enough on its own? Explain your prediction.

3. What evidence would convince you that resonance - not just loudness - is responsible?

DURING - collect evidence

1. List every variable the team changes or controls.

2. What evidence shows that the glass itself is vibrating before it breaks?

3. Identify one fair-test strength and one weakness in the method.

1. Breaking Glass - continued

AFTER - explain and evaluate

1. Use the words natural frequency, resonance, amplitude and energy to explain the result.

2. Why would two glasses that look similar not necessarily respond to the same note?

3. Design a safer school investigation that demonstrates resonance without breaking glass.

TRANSFER - new situation

A loudspeaker plays a note close to an object's natural frequency. Predict what happens as the driving frequency is moved gradually away from resonance, and justify your prediction.

Final judgement: What does the evidence support? What does it NOT prove? Give one improvement to the test.

2. Voice Flame Extinguisher

Season 5, Episode 7 (2007) | **Physics:** Sound as a pressure wave, amplitude, energy and interaction with matter

The episode tests whether sound can disturb air enough to extinguish a flame, making pressure variations and sound energy visible through an effect.

BEFORE - predict

1. Can sound move matter even though the sound wave does not carry the air across the room?

2. Which should matter more here: frequency, amplitude, distance, or all three? Rank them and justify.

3. What measurement would make the test more scientific than simply saying the sound is “loud”?

DURING - collect evidence

1. What changes between the unamplified and amplified tests?

2. What observations support the idea that sound consists of pressure variations in air?

3. Does the result prove that any loud sound can extinguish any flame? Explain.

2. Voice Flame Extinguisher - continued

AFTER - explain and evaluate

1. Explain how a sound wave can transfer energy without transferring the air from source to flame.

2. Suggest two variables that should be controlled in a repeat investigation.

3. Distinguish the evidence from the MythBusters' conclusion.

TRANSFER - new situation

Predict how moving the sound source twice as far from the flame would affect the result. State what extra evidence you would want before being confident.

Final judgement: What does the evidence support? What does it NOT prove? Give one improvement to the test.

3. Deep Space Hollywood

MythBusters Jr. - Season 1, Episode 8 (2019) | **Physics:** Sound needs a medium; particle model and sound in space

One myth asks whether anyone in space can hear you scream - a direct route into why mechanical sound waves need particles.

BEFORE - predict

1. If an astronaut shouts outside a spacecraft, what happens to the vocal cords? What happens to the sound?

2. Why can light cross space but ordinary sound cannot?

3. Draw a particle diagram for sound travelling through air and another for a vacuum.

DURING - collect evidence

1. Which part of the demonstration models the absence of a medium?

2. What observation is evidence that sound transmission changes when particles are removed?

3. What alternative explanation would you need to rule out?

3. Deep Space Hollywood - continued

AFTER - explain and evaluate

1. Explain the result using vibrations and collisions between particles.

2. Why can astronauts communicate by radio even though they cannot hear each other directly through space?

3. What limitation does a classroom vacuum demonstration have compared with actual space?

TRANSFER - new situation

Imagine a bell inside a sealed jar as the air is slowly removed. Sketch and explain how the detected sound amplitude should change.

Final judgement: What does the evidence support? What does it NOT prove? Give one improvement to the test.
